Dataset Profile EDA Report

Raw Data Profiling Before VPR Experiments

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April 24, 2026

Purpose

This report focuses on **raw dataset profiling before any descriptor experiments**. The goal is to understand what the datasets already tell us on their own:

- how large they are,
- whether the splits are balanced,
- whether day and night images follow similar distributions,
- whether the campus query set is dominated by hard no-match cases, and
- whether the campus walk is evenly covered across places.

The profiling code is in `analysis/dataset_profile_eda.py`, and the generated plots are saved under `output_images/eda/data_profiling/`.

Question 1: Are the datasets balanced at the split level?

Answer: GardensPoint is balanced, but campus is smaller and less balanced.

Dataset	Database / Day	Query / Night
GardensPoint	200	200
Campus strict	50	64

Interpretation.

- GardensPoint behaves like a clean benchmark split.
- The campus dataset is much smaller, so each difficult example has a larger effect on the final metrics.
- This matters experimentally because variance will be lower on GardensPoint and much higher on campus. A small gain or drop on campus should therefore be read more cautiously, especially when only a few hard queries can move the metric noticeably.

Question 2: Are image sizes and aspect ratios consistent?

Answer: GardensPoint day and night images do **not** share the same resolution, while the campus data is size-consistent.

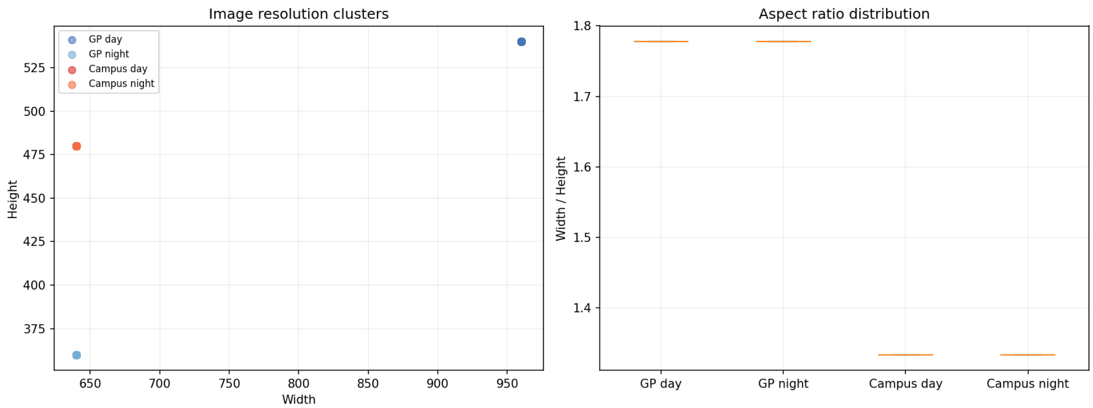

Figure 1: Resolution and aspect-ratio profiling before any descriptor extraction.

- GardensPoint day images average **960x540**.
- GardensPoint night images average **640x360**.
- Campus day and night images both average **640x480**.

Interpretation. This means the benchmark already contains a raw size mismatch between day and night, while campus difficulty comes from content and illumination rather than image size. In practice, resizing inside the descriptor pipeline may hide this difference partially, but the EDA still tells us that GardensPoint and campus are stressing the model in different ways before any features are extracted.

Question 3: How different are the raw day and night image distributions?

Answer: The two datasets show **different kinds of night-time shift**.

Split	Brightness Mean	Contrast Mean	Sharpness Mean
GardensPoint day	97.1	58.9	405.8
GardensPoint night	127.6	69.8	718.3
Campus day	112.0	53.5	784.2
Campus night	79.1	48.4	719.7

Interpretation.

- On GardensPoint, night images are **brighter**, higher-contrast, and sharper than day images.
- On campus, night images are **darker** and slightly lower in contrast.
- Visually, the GardensPoint night frames appear **exposure-compensated or over-brightened**, rather than simply representing a naturally dark night condition. The EDA is consistent with that impression.

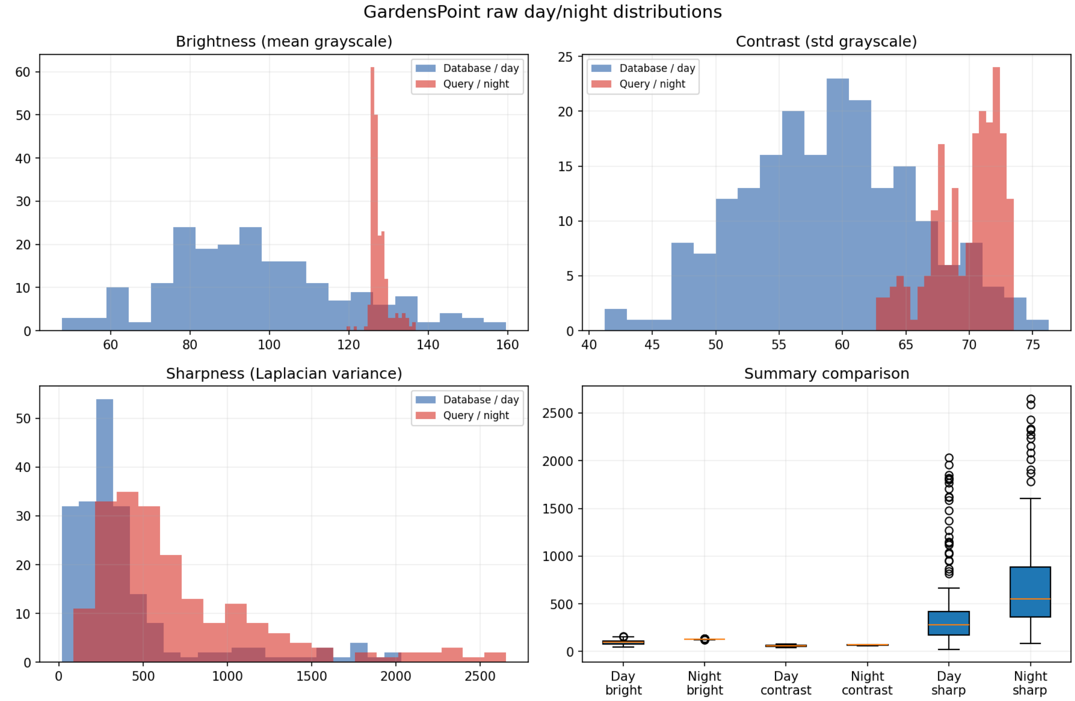

Figure 2: Raw day versus night distributions on GardensPoint.

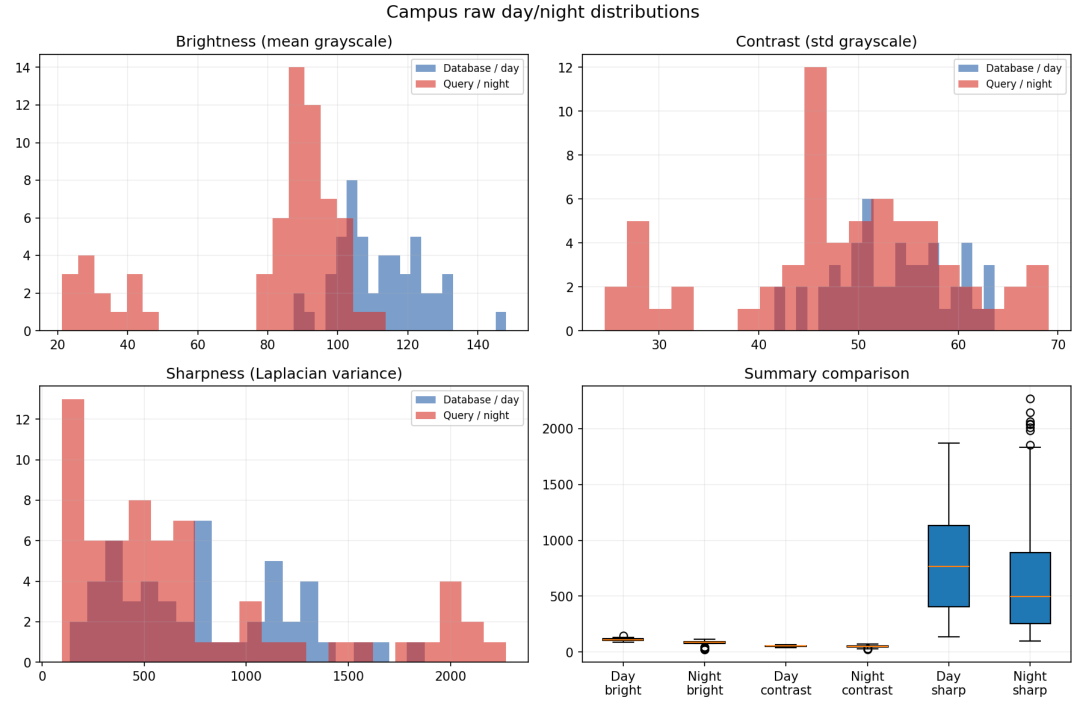

Figure 3: Raw day versus night distributions on the campus dataset.

- So GardensPoint and campus do not represent the same visual night condition. This is one reason why a descriptor can look strong on GardensPoint and still struggle to transfer fully to campus.

- Conceptually, this is why raw-distribution EDA matters: if the benchmark night split is artificially brightened, then benchmark success may reflect robustness to viewpoint and route alignment more than robustness to truly dark night imagery.

Question 4: How challenging is the strict campus query composition before any model is used?

Answer: The campus strict query set is hard even before descriptor extraction because many queries are not simple one-to-one matches.

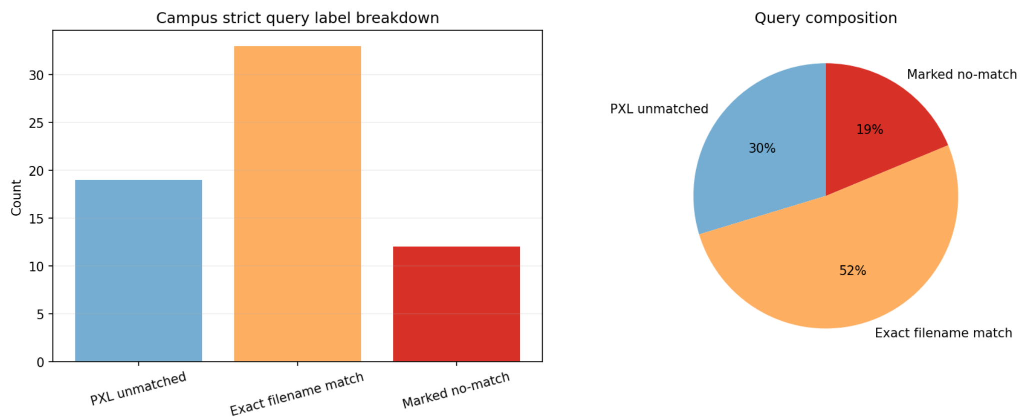

Figure 4: Strict campus query composition before experimentation.

The strict campus query split contains:

- **33** exact filename matches,
- **12** marked no-match queries, and
- **19** `PXL_*` phone images with no strict day counterpart.

Interpretation. Nearly half the strict campus query set is effectively a hard rejection or non-one-to-one retrieval problem. This means low precision-style metrics on campus are not only a descriptor issue; they are also a dataset-structure issue. These `PXL_*` images are still part of the strict dataset and are intentionally kept as hard no-match queries. So before running any model, the EDA already warns us that thresholded matching will be much harder than top-k retrieval.

Question 5: Is place coverage balanced across the campus walk?

Answer: No. The place-level annotations show clear imbalance.

Main observations.

- There are **10** unique place IDs.
- The largest night bucket is `entrance` with **25** images.

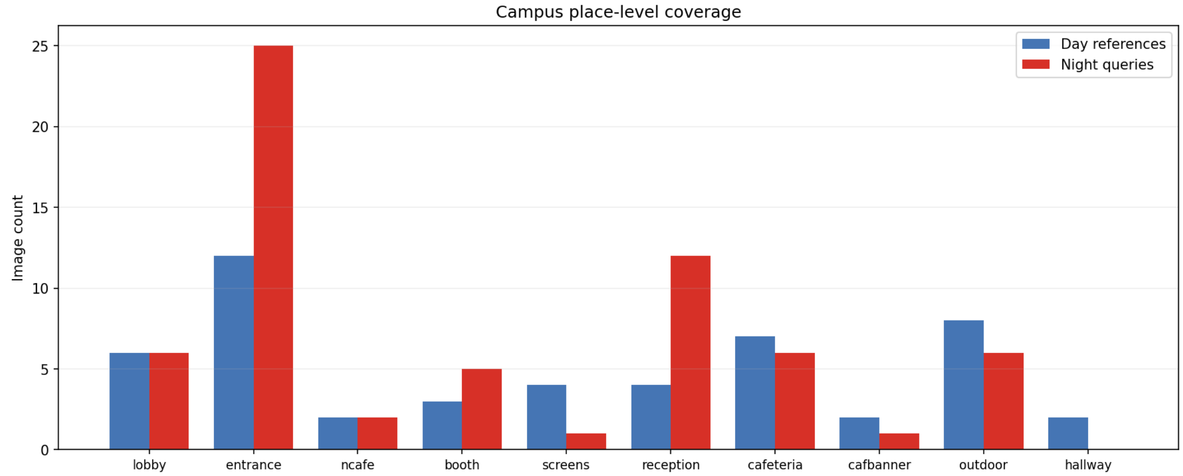

Figure 5: Place coverage across the campus walk.

- Some locations have very limited support, and `hallway` has no night counterpart in the rematched place-level flow.

Interpretation. The campus walk is not uniformly sampled, so some regions will dominate retrieval behavior more than others. In practice, descriptors will see many more examples of `entrance` than of smaller places such as `ncafe` or `cafbanner`. That means a model can appear strong simply because the most common places are easier to retrieve, while underrepresented places may stay fragile. This is why the place-coverage EDA matters before experimentation: it tells us that raw dataset frequency can bias which scenes dominate the final metrics.

Question 6: What do representative samples look like?

Answer: The montages confirm the statistical findings visually.

Figure 6: Representative GardensPoint day and night samples.

Interpretation.

- GardensPoint shows cleaner route progression and visually consistent nighttime lighting, but the night frames also look exposure-lifted, which helps explain why their average brightness is unexpectedly higher than the day split.

Figure 7: Representative campus day and night samples.

- Campus shows stronger viewpoint variation, more uneven lighting, and more heterogeneous scenes, so it is a better stress test for transfer but a noisier benchmark for clean descriptor comparison.

Conclusion

This profile EDA shows that the raw data itself already explains part of our later experimental behavior:

- GardensPoint is a useful controlled benchmark, but its night distribution is not identical to campus.
- The campus strict split is difficult before any model is even chosen because of its no-match composition.
- The campus walk is also unevenly sampled across places.

So the raw data supports our experimental design: use GardensPoint as the controlled benchmark, use campus as the harder transfer test, and interpret campus results with an awareness of label composition and place imbalance. More importantly, this EDA gives us a checklist for reading later experiments correctly: benchmark gains do not automatically imply campus robustness, precision-style failures may come from query composition rather than descriptor weakness alone, and strong scores on common places do not guarantee balanced performance across the whole route.